

NAIROBI SCHOOL 2025 PREDICTION



KCSE ANTICIPATION EXAM



231/1

BIOLOGY

PAPER 1 (THEORY)

TIME: 2 HOURS

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

(Kenya Certificate of Secondary Education)

INSTRUCTIONS TO THE CANDIDATES

- Write your name and admission number in the spaces provided above
- Write the name of your school and the date of examination in the spaces provided above
- Answer **ALL** the questions in the spaces provided.
- Answers must be written in the spaces provided in the question paper.
- Additional papers must not be inserted.
- Candidates should answer the questions in English.

FOR EXAMINER'S USE ONLY

QUESTION	MAXIMUM SCORE	CANDIDATE'S SCORE
1-31	80	

Answer **ALL** the questions in the spaces provided.

1. Two students were observing bacteria using identical microscopes and identical slides. Student A saw 10 bacteria and student B saw 50 bacteria.

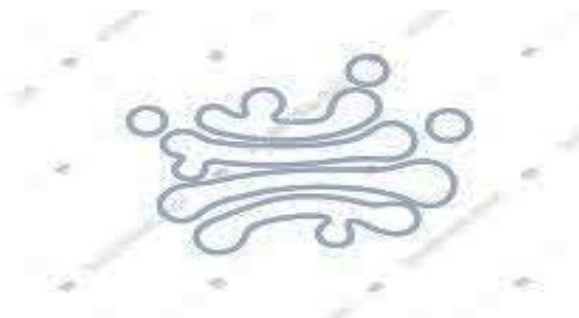
a) Suggest a reason why they observed different number of bacteria. (1 mark)

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b) Which of the following combination would give a wider field of view, eye piece X10 and objective lens X20 or eye piece X10 and objective lens X40? (1 mark)

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2. The diagram below represents a cell organelle.



i) Identify the organelle (1 mark)

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ii) State the function of the organelle (1 mark)

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3. A plastic bottle full of water was stoppered using a piece of stem obtained from a young herbaceous plant whose epidermis had been peeled off. After 24 hours, it was noted that the stopper closed the container tightly.

a) Why was the epidermis peeled? (1 mark)

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b) Account for the observation made (2 marks)

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4. Give the roles of the following in the alimentary canal:

a) Hydrochloric acid in the stomach (1 mark)

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b) Mucus (1 mark)

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c) Name the cells that secrete enzyme pepsin and renin in the stomach (1 mark)

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d) Enzyme enterokinase activates trypsinogen to trypsin. Give the origin of the enzyme (1 mark)

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5. Give the roles of the following hormones: (2 marks)

i) Secretin

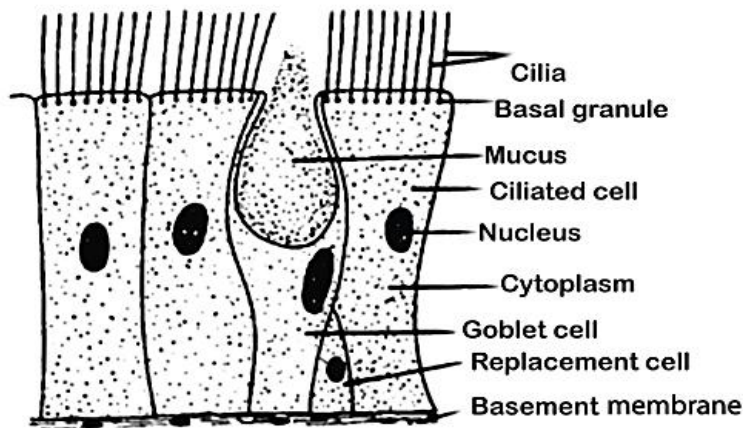
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ii) Cholecystokinin

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6. The figure below shows some cells that form part of the mammalian respiratory tract.



a) Identify the part of the respiratory tract (1 mark)

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b) Give the role of the: (2 marks)

i) Goblet cell

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ii) Cilia

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7. Alveoli do not collapse when we breath out. Explain (1 mark)

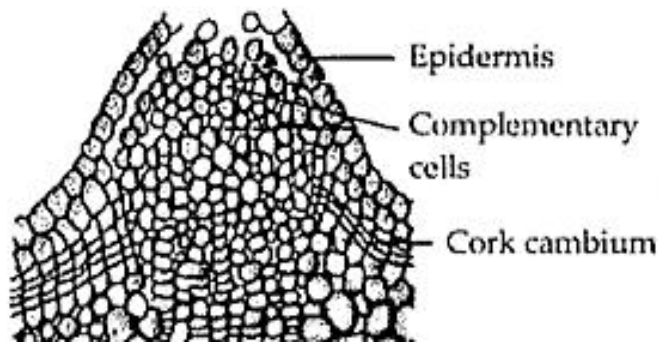
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8. Name the site in the human body where parallel flow is operational. (1 mark)

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9. The diagram below shows the structure of gaseous exchange in a woody plant



i) Identify the gas that diffuse (2 marks)

In _____

Out _____

ii) Give two adaptations of the structure (2 marks)

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10. Explain how sunken stomata affect the rate of transpiration. (2 marks)

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11. Give two adaptations of the liver to its functions (2 marks)

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b) Explain why insulin is not administered orally. (1 mark)

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12. Give the role of the following hormones in reproduction: (3 marks)

a) Prolactin

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b) Oxytocin

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c) Relaxin

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13. What is blood transfusion? (1 mark)

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b) state one factor to consider during blood transfusion (1 mark)

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14. Give reasons for the following:

i) Donated blood should be discarded after 21 days (1 mark)

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ii) Upon donation, the blood donor is usually given a drink (1 mark)

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15. A transfusion of Rh+ blood was given to a patient Rh- blood. After a week, a similar transfusion was given to the same patient. What is likely to be the effect of the second transfusion?
(2 marks)

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16. Name the form in which:

i) Carbon (IV) oxide is transported in blood.

(1 mark)

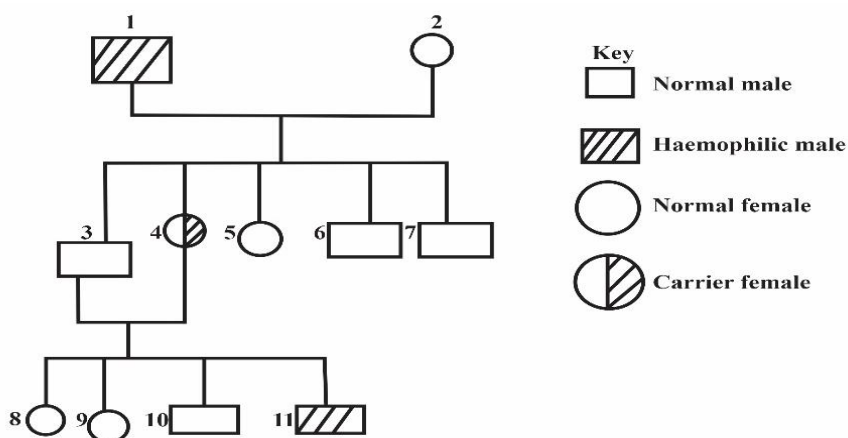
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ii) Give the role of the enzyme found in the red blood cells

(1 mark)

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17. Haemophilia is a sex-linked condition. The following pedigree shows a portion of a family in which members are haemophiliac. Use H to represent the gene for haemophilia and h to represent the gene for non-haemophilia.



a) Identify the genotypes of the parents:

1 _____ 2 _____

(2 marks)

b) What is the genotype of the offspring 4

(1 mark)

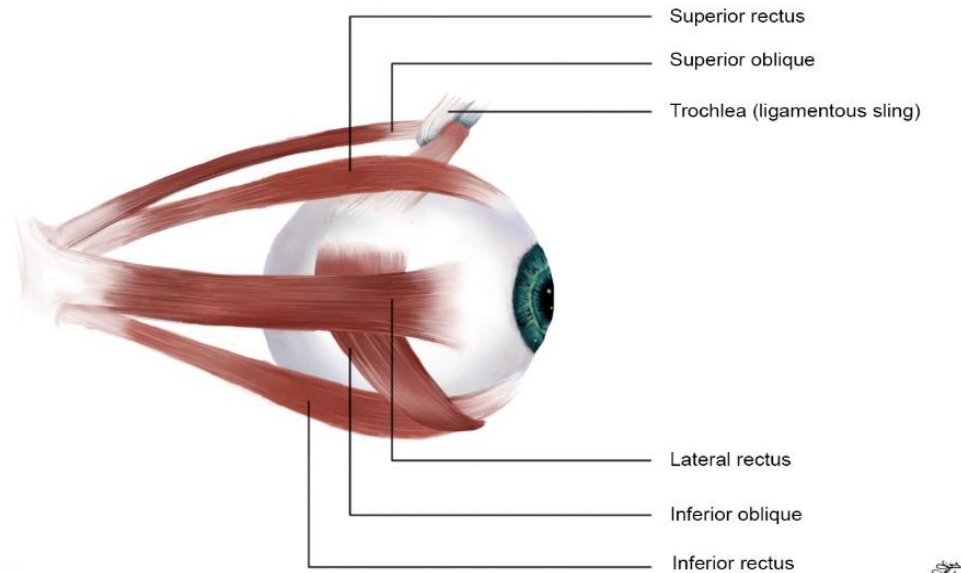
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c) Explain the prevalence of haemophilia in males compared to females

(2 marks)

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18. The diagram below represents the mammalian eye.



a) Give the role of the muscles:

(3 marks)

i) Superior and inferior rectus muscles

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ii) Lateral rectus muscles

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iii) Oblique muscles

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b) What is accommodation of the eye?

(1 mark)

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c) Describe accommodation of close objects

(3 marks)

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19. Give one difference between aerobic and anaerobic respiration **(1 mark)**

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b) Give two importance of respiratory quotient **(2 marks)**

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20. Explain why there is increased heartbeat during vigorous exercise in man. **(1 mark)**

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21. What is eutrophication? **(1 mark)**

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b) Give two effects of eutrophication. **(2 marks)**

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22. Name the organisms involved in the following processes. **(2 marks)**

i) Nitrification _____

ii) Denitrification _____

23. Give two functions of cerebral spinal fluid **(2 marks)**

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24. Which parts of the brain are concerned principally with:

- i) controlling body temperature _____ (1 mark)
- ii) interpreting what is seen _____ (1 mark)

25. Name the structure responsible for maintaining posture and balance in the mammalian ear
(1 mark)

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26. What is the disadvantage of the eustachian tube? (1 mark)

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27. The diagram below shows the leaves of *Medicago marina* during the day B, and at night A.



- a) Name the response _____ (1 mark)
- b) State the survival value of the response (2 marks)

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28. Name the cartilage found in between the bones of the vertebral column (1 mark)

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- b) Give two functions of the cartilage named above (2 marks)

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- 29.** State the type of muscle fiber found in the following regions: **(3 marks)**
- a) Stomach _____
 - b) Aorta _____
 - c) Left ventricle _____

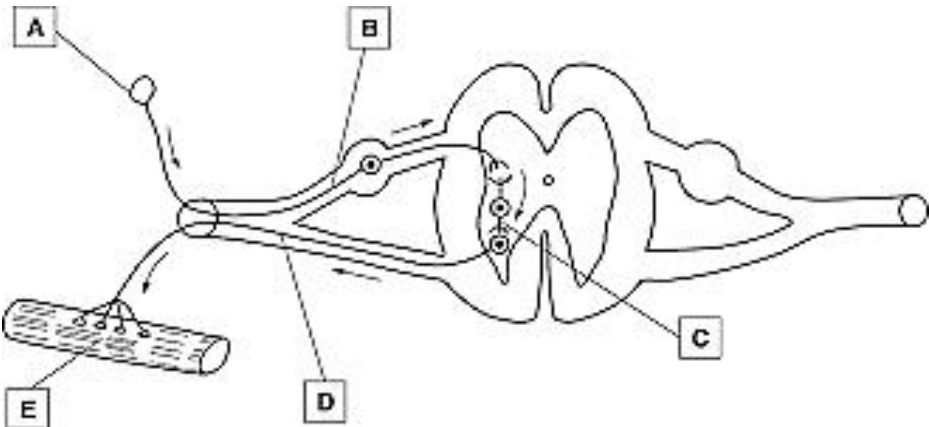
- 30.** Which part of the flower develops into the following structures after fertilization: **(2 marks)**
- a) Seed coat _____
 - b) Fruit _____

31. What is a reflex arc? **(1 mark)**

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b) The figure below outlines the basic components of a reflex arc in humans



a) Name the type of neurone labeled C **(1 mark)**

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b) Name the part of the arc labeled A and E **(2 marks)**

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